

CITYTWIN · DESIGN DECISIONS REGISTER

CityTwin design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CityTwin design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, CTW-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Decisions still to be made.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Sun shield standoff holes in the cabinet sides	(a) drill only kiosks that get the shield; (b) drill every cabinet, with sealed blanking screws until a shield is fitted	(a)	Cabinet drilling (build plan section 3.3)	CTW-DDR-003, A2
2	First partner city or neighborhood for co-design and a pilot site; this also decides whether the frost heater is fitted	Not yet listed	None made	Site, footing and climate rules; whether the outdoor parts are built	CTW-DDR-001, O1; CTW-DDR-002
3	Antenna height needed for the reference neighborhood with the gateway in the cabinet	Post-top antenna at 2.38 m, as modeled; a taller mast; a rooftop gateway	None made	Antenna bracket and coax length	CTW-DDR-002, Q1
4	Languages and scripts on the kiosk, and how residents ask questions or report a fault	For co-design with the partner city (decision 2)	None made	Notice plate content; screen layouts	CTW-DDR-002, Q2
5	Street climate (R20): the panel still exceeds its 0 to 40 °C rating at 45 °C air even in shade	A wider-range panel if one exists; a heated, insulated panel bay for a frost city; shaded siting only	None made beyond the site rules of CTW-DDR-002 N4	Head interior; not part of the pilot build	CTW-PRC-001, open questions
6	Pull paths for PotholeLog segment files and DockHub hourly aggregates: formats and hosts	To be agreed with those projects	Adopt the outbound pull (decided, D11); formats still to agree	Software only	CTW-DDR-001, D11
7	A LoRaWAN uplink from CrossSafe, whose radio now runs point to point (R1)	To be defined in CrossSafe	None made in this repo	Software only	CTW-REQ-001, R1

■ To confirm when parts are bought

Table 2. Items to confirm against the chosen part.

#	What to confirm	Why it matters	Source
1	The cabinet comes with a removable mounting plate of about 340 x 390 mm on studs that stand it at least 10 mm off the back wall, and a lockable door on the 360 x 460 mm face	The rails, the cabinet screws behind the plate and the 25 mm cable hole above the plate are placed for this	CTW-DDR-003, P9
2	The e-paper panel's outline (210 x 290 x 7 mm assumed) and where its flat cable leaves the panel	Sets the foam gasket, the carrier's ribbon slot and the driver board position	CTW-DDR-003, P4
3	The M8 steel flush rivet nuts' grip range covers the 4 mm post wall with its coating, and their countersink size	The head and cabinet screws depend on them	CTW-DDR-003, P1
4	The buttons' LED rings run from 5 V or 12 V, and their nut size fits beside the notice plate	Sets the LED driver wiring and the clearance below the notice plate	CTW-DDR-003, P5, P6
5	TwinKit fits a backup pack of at least 1.84 Ah for cabinet installs (now 1.5 Ah); until then R18 is not met in the worst case	The outage ride-through of 2 h	CTW-DDR-002, N3
6	The hood light's LED strip and dusk sensor work from the 12 V supply through the LED driver, dimmed to about 0.3 W	Night lighting (R9) and running power (R15)	CTW-DDR-002, N6
7	The anchor design resistance (10 kN per M16 anchor assumed) and the footing, from a structural design to the local code	Wind (R13) and the safety of a kiosk of about 64 kg beside a walkway	CTW-CAL-001, section H

■ Value engineering

Value-engineering target: USD 750 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 778 for the pilot kiosk (USD 28 over the target); USD 808 with the outdoor sun shield, and USD 1,068 with the TwinKit gateway, which is costed in TwinKit.

- **Main cost drivers:** the parts added to make the kiosk buildable (display carrier USD 8, fixings USD 22, and USD 13 more for the head's converter, LED driver and terminal block, less USD 4 moved out of another item); the outdoor sun shield (USD 30), reported separately.
- **Savings worth trying:** a lower-cost cabinet; dropping the real-time clock; checking prices at purchase, since the figures are indicative.

■ Decisions made

Table 3. Decisions made, with the record that argues each one.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: budget covers the kiosk only; color e-paper with an indoor LCD variant; sheltered pilot; gateway in the kiosk cabinet; mains power; 5 per hour small-count threshold and two years of raw data; CC BY 4.0; buttons; 46-node reference neighborhood; one-way publishing; outbound pulls for PotholeLog and DockHub	Amish: "i accept all your recommendations, go with them across all repos."	CTW- DDR-001, CTW- DDR-002
2026-09-25	N1 to N6: budget_usd \$750; R10 restated (LED ring within 0.5 s, layer within 25 s); larger TwinKit pack requested; cabinet sun shield and shaded screen as outdoor site rules, heater only for a frost city; pilot range 0 to 35 °C air; hood light dimmed to about 0.3 W	Amish, same instruction	CTW- DDR-002
2026-10-01	Design for construction, P1 to P14: rivet nuts in the post for the head and cabinet; folded head tray with a removable front panel; window bonded with glazing tape; display carrier on studs; 5 V converter, LED driver and terminal block in the head; button, notice plate and hood fixings; mounting plate in the cabinet; grommeted cable holes; conduit hole and gland; vent hole for galvanizing; antenna bracket plate; shield on standoffs with a thumb-screw back panel	Made under Amish's 2026-09-30 instruction to make the design physically buildable ("If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."); open for his review	CTW- DDR-003